Radiotrophic Human Cell Metabolism — Project Plan

Paper Title (working)

"Computational Feasibility Analysis of Melanin-Based Radiotrophic Metabolism in Engineered Human Cells"

Core Question

What is the minimum viable gene set to establish functional radiotrophic metabolism in a human cell, and does the energy-to-oxidative-damage tradeoff produce a net benefit?

WHAT WE HAVE (Phase 1 — COMPLETE)

Custom Metabolic Model (COBRApy)

- 50-reaction, 50-metabolite constraint-based model
- Glycolysis, TCA cycle, electron transport chain
- Melanin synthesis pathway
- Radiotrophic NADH generation reaction
- Four ROS defense systems: native (SOD, catalase, GPX, GR) + engineered (Dsup, Mn-AOX, Nrf2)
- File: `radiotrophic_model.py`

Key Results

1. **+47% ATP at baseline** — radiotrophic cell produces 178.6 vs 121.4 ATP at full nutrients
2. **+781% ATP at glucose=0.5** — advantage scales inversely with nutrient availability
3. **125 ATP from radiation alone** at zero glucose (normal cell = 0)
4. **ROS cost is manageable** — SOD handles radiotrophic superoxide; engineered defenses are redundant when SOD is intact
5. **Hypoxia limits the benefit** — radiotrophic NADH still needs O2 for ETC, so benefit drops under hypoxia
6. **Linear dose-response** — each unit of radiotrophic flux yields ~1.14 ATP net after ROS cost

Known Limitations

- ROS coefficient (0.3 superoxide per NADH) is assumed, not literature-grounded

- Radiotrophic flux cap of 50 is arbitrary
- Perfect melanin-to-ETC electron coupling is assumed
- No kinetic validation (FBA = steady-state only)
- Custom model avoids Recon3D's thermodynamic infeasibility loops but loses genome-scale resolution

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WHAT WE NEED TO DO

Phase 2: Ground the Stoichiometry (1–2 sessions)

Goal: Replace assumed coefficients with literature-derived values

Tasks:

- [ ] Search Dadachova et al. 2007 (PLOS ONE) for measured electron transfer rates in melanized C. neoformans under gamma radiation
- [ ] Search Dadachova & Casadevall 2008 (Current Opinion in Microbiology) for radiosynthesis efficiency estimates
- [ ] Search Robertson et al. 2012 for melanin quantum yield under ionizing radiation
- [ ] Extract: NADH yield per Gray of radiation, ROS generation rate, melanin degradation rate
- [ ] Update model stoichiometry with grounded values
- [ ] Re-run all experiments with corrected coefficients
- [ ] Document parameter sources in a supplementary table

Phase 3: COPASI Kinetic Modeling (2–3 sessions)

Goal: Validate that the steady-state FBA results hold dynamically

Tasks:

- [ ] Download COPASI (standalone GUI app — copasi.org)
- [ ] Build ODE model of the radiotrophic + ROS defense subsystem
- [ ] Key reactions: RADIO -> NADH + ROS, SOD scavenging, catalase, Dsup
- [ ] Simulate radiation pulse: what happens to ROS levels transiently before SOD catches up?
- [ ] Find: maximum tolerable radiotrophic flux before transient ROS exceeds lethal threshold
- [ ] Generate time-course plots (ROS, NADH, ATP over time after radiation exposure)
- [ ] Compare steady-state predictions (COBRApy) vs dynamic behavior (COPASI)

Phase 4: AlphaFold Structural Analysis (1 session)

Goal: Assess structural feasibility of Dsup functioning in human cells

Tasks:

- [ ] Go to ColabFold (colab.research.google.com — free, no install needed)
- [ ] Get tardigrade Dsup sequence from UniProt (Ramazzottius varieornatus, Q9GV38 or similar)
- [ ] Run AlphaFold2 structure prediction
- [ ] Compare predicted Dsup structure with human histone complex (PDB: 1KX5)
- [ ] Assess: does Dsup's DNA-binding surface remain accessible in predicted fold?
- [ ] Generate structural figures for the paper

Phase 5: Write the Paper (3–4 sessions)

Goal: Complete manuscript draft

Proposed structure:

1. **Abstract** — one-paragraph summary of question, methods, results
2. **Introduction** — radiotrophic fungi, cross-species gene transfer precedent (Dsup in HEK293), gap in literature
3. **Methods**
   - 3.1 Custom metabolic model construction
   - 3.2 Flux balance analysis experimental design
   - 3.3 ODE kinetic model (COPASI)
   - 3.4 Structural feasibility (AlphaFold)
4. **Results**
   - 4.1 Radiotrophic ATP supplementation under nutrient restriction
   - 4.2 Radiation dose-response and ROS tradeoff
   - 4.3 Defense system ablation analysis
   - 4.4 Dynamic ROS kinetics validation
   - 4.5 Dsup structural compatibility
5. **Discussion** — biological feasibility, engineering challenges, applications (space biology, radiation medicine), limitations
6. **Supplementary** — full model specification, parameter table, interactive 3D visualization

Target journals (in order):

1. Synthetic Biology (Oxford) — perfect fit, computational + synthetic biology
2. BMC Systems Biology
3. PLOS Computational Biology

4. Frontiers in Bioengineering and Biotechnology

FILE INVENTORY

```
radiotrophic_project/
├── radiotrophic_model.py          # Complete model + all experiments
├── radiotrophic_cell_model.json   # Saved COBRApy model (JSON)
├── PROJECT_PLAN.md                # This file
├── results/                       # Generated CSV files from experiments
│   ├── glucose_restriction.csv
│   ├── hypoxia.csv
│   ├── dose_response.csv
│   ├── ablation.csv
│   └── combined_stress.csv
└── (future)
    ├── copasi/                    # COPASI kinetic models
    ├── alphafold/                 # Structural analysis
    ├── figures/                   # Publication figures
    └── manuscript/                # Paper drafts
```

SETUP INSTRUCTIONS (Windows)

```bash
# Install Python packages
pip install cobra pandas matplotlib

# Download Recon3D (for reference, not used in custom model)
# http://bigg.ucsd.edu/static/models/Recon3D.xml.gz

# Run all experiments
cd radiotrophic_project
python radiotrophic_model.py

# Download COPASI for Phase 3
# https://copasi.org/Download/

# AlphaFold via ColabFold for Phase 4
# https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.
```